Social Media Behavior & Psychological Impact Analysis

Data Analysis Team

April 9, 2026

1 Project Overview

This project aims to analyze the relationship between social media usage behavior and its potential psychological impact.

We use two datasets:

- **Dataset 1 (Kaggle):** Psychological and behavioral dataset including anxiety, burnout, wellbeing, and lifestyle factors.
- **Dataset 2 (Survey):** Behavioral dataset capturing platform usage, content type, sleep patterns, and doomscrolling behavior.

Core Idea: Use behavioral patterns (Dataset 2) to explain or predict psychological outcomes (Dataset 1).

2 Dataset Description

2.1 Dataset 1 (Psychological Dataset)

Type: Behavioral + Psychological Outcomes

Contains:

- Anxiety, Burnout, Wellbeing
- Screen Time, Social Media Usage
- Sleep, Exercise, Caffeine

Limitations:

- No platform-level detail
- Self-reported scores

2.2 Dataset 2 (Behavioral Dataset)

Type: Behavioral + Content Usage

Contains:

- Platform usage (TikTok, Instagram, etc.)
- Doomscrolling behavior
- Sleep patterns
- Content type and format

Limitations:

- No psychological measurements
- Limited to age group 18–24

Note: Some variables in Dataset 2 are categorical (e.g. screen time ranges, sleep ranges) and require preprocessing before quantitative analysis.

3 Analytical Framework

The analysis will be conducted on three levels:

- **Level 1: Behavioral Analysis (Dataset 2)** Understanding user behavior patterns.
- **Level 2: Psychological Analysis (Dataset 1)** Understanding mental health outcomes.
- **Level 3: Cross-Dataset Insights** Linking behavior to psychological impact.

Note: The two datasets are not merged at the individual level. Cross-dataset analysis is based on pattern comparison and hypothesis generation.

4 Key Analytical Questions

4.1 Dataset 1 (Psychological)

- **Is screen time associated with higher anxiety levels?**
  Columns: Daily_Social_Media_Hours, Screen_Time_Hours, Anxiety_Score

- **Does sleep duration affect burnout risk?**
  Columns: Daily_Sleep_Hours, Sleep_Quality_Score, Burnout_Risk

- **Does exercise reduce emotional fatigue?**
  Columns: Exercise_Frequency_per_Week, Emotional_Fatigue_Score

- **Which group shows the highest burnout levels?**
  Columns: Burnout_Risk, Student_Working_Status, Gender, Country

- **Is there a threshold of screen time where wellbeing drops significantly?**
  Columns: Screen_Time_Hours, Wellbeing_Index

4.2 Dataset 2 (Behavioral)

- **Which platform is most associated with doomscrolling?**
  Columns: top_apps, doomscroll_category, high_doomscroll_low_sleep

- **Does higher screen time lead to lower sleep duration?**
  Columns: daily_screen_time_hours, sleep_hour, sleep_level

- **Do TikTok users have worse sleep patterns than others?**
  Columns: top_apps, sleep_hour, sleep_level

- **Is morning phone checking linked to irritability?**
  Columns: check_phone_morning, irritable_recently

- **Does content type influence usage intensity?**
  Columns: ContentFormat_ID, screen_time_level, social_media_level

4.3 Cross-Dataset Questions

- **Does doomscrolling behavior lead to higher anxiety?**
  DS2 Columns: doomscroll_category, high_doomscroll_low_sleep
  DS1 Columns: Anxiety_Score, Social_Comparison_Index

- **Do high screen time and low sleep increase burnout risk?**
  DS2 Columns: high_screen_low_sleep
  DS1 Columns: Burnout_Risk, Sleep_Risk

- **Does content type influence behavioral patterns and psychological outcomes?**

DS2 Columns: ContentFormat_ID, StayFactor_ID

DS1 Columns: Content_Type_Preference, Anxiety_Score

5 Dashboard Questions

Note: Each dashboard should include filters for Gender, Age Group, and Usage Level where applicable.

5.1 Dataset 1 Dashboard (Psychological)

- **How do anxiety, burnout, and wellbeing vary across demographics?**
  Columns: Anxiety_Score, Burnout_Risk, Wellbeing_Index, Age_Group, Gender, Country

- **What is the relationship between screen time and psychological outcomes?**
  Columns: Screen_Time_Hours, Anxiety_Score, Mood_Stability_Score, Burnout_Risk

- **How does sleep impact wellbeing and productivity?**
  Columns: Daily_Sleep_Hours, Sleep_Quality_Score, Wellbeing_Index, Productivity_Risk

- **What are the characteristics of high-risk users?**
  Columns: Burnout_Risk, Sleep_Risk, Emotional_Risk, Social_Comparison_Index

- **How do lifestyle factors influence mental health?**
  Columns: Exercise_Frequency_per_Week, Caffeine_Intake_Cups, Anxiety_Score, Emotional_Fatigue_Score

5.2 Dataset 2 Dashboard (Behavioral)

- **What is the distribution of platform usage across users?**
  Columns: top_apps, gender, status

- **How does screen time relate to sleep patterns?**
  Columns: daily_screen_time_hours, sleep_hour, sleep_level

- **Which platforms are most associated with doomscrolling behavior?**
  Columns: top_apps, doomscroll_category, high_doomscroll_low_sleep

- **How does content format influence user engagement and usage intensity?**
  Columns: ContentFormat_ID, screen_time_level, social_media_level, StayFactor_ID

- **What behavioral patterns are linked to irritability?**
  Columns: irritable_recently, check_phone_morning, daily_screen_time_hours

5.3 Combined Dashboard

- **How does screen time relate to anxiety and burnout risk?**
  DS1 Columns: Screen_Time_Hours, Anxiety_Score, Burnout_Risk
- **How does sleep behavior impact psychological wellbeing?**
  DS1 Columns: Daily_Sleep_Hours, Wellbeing_Index, Sleep_Risk
- **What behavioral patterns are associated with high-risk users?**
  DS2 Columns: high_screen_low_sleep, doomscroll_category
  DS1 Columns: Burnout_Risk, Emotional_Risk
- **How does content consumption behavior relate to psychological outcomes?**
  DS2 Columns: ContentFormat_ID, top_apps
  DS1 Columns: Content_Type_Preference, Anxiety_Score

6 Key Performance Indicators (KPIs)

- **Psychological Risk Index**
  Columns: Anxiety_Score, Burnout_Risk, Sleep_Risk, Overthinking_Score
  Definition: Composite indicator of overall psychological risk
  Note: Requires normalization of component variables before aggregation
- **Digital Dependency Score**
  Columns: Screen_Time_Hours, Daily_Social_Media_Hours, Exercise_Frequency_per_Week, Daily_Sleep_Hours
  Definition: Measures dependence on digital usage relative to healthy habits
- **Burnout Rate**
  Columns: Burnout_Risk
  Definition: Percentage of users classified as high burnout
- **Doomscroll Rate**
  Columns: doomscroll_category, high_doomscroll_low_sleep
  Definition: Percentage of users exhibiting doomscrolling behavior

- **Sleep Risk Percentage**
  Columns: sleep_level, high_screen_low_sleep
  Definition: Percentage of users with unhealthy sleep patterns

- **High Screen Usage Rate**
  Columns: screen_time_level, daily_screen_time_hours
  Definition: Percentage of users with high or very high screen time

- **Content Engagement Index**
  Columns: ContentFormat_ID, StayFactor_ID, SkipPoint_KEY
  Definition: Measures engagement level based on content type and retention factors

7 Open Questions (Brainstorming)

- Can we identify user personas based on behavior and outcomes?
- Is there an addiction loop (short content → doomscrolling → poor sleep)?
- Are there healthy usage patterns that protect mental wellbeing?

8 Conclusion

This project aims to bridge the gap between behavioral data and psychological outcomes, enabling deeper insights into digital habits and their potential impact on mental health.